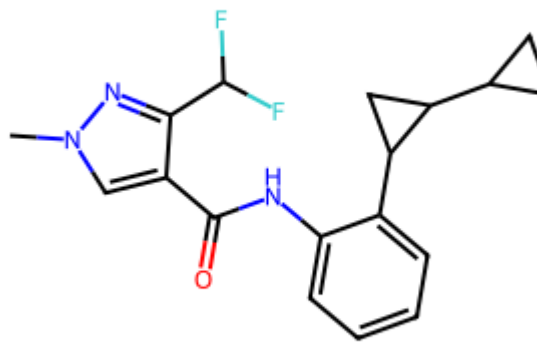
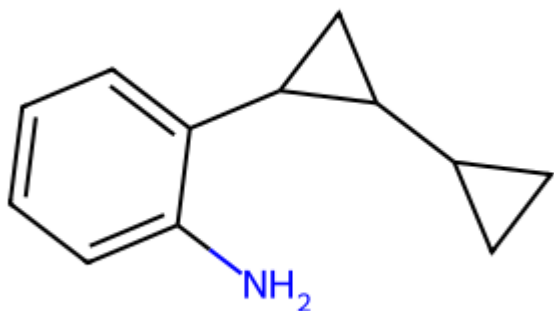




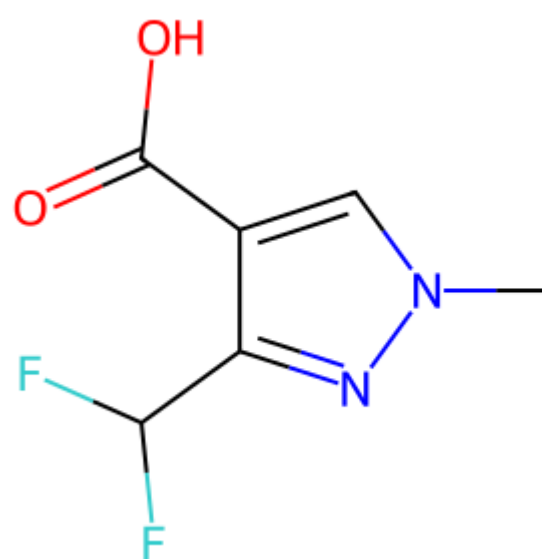
{'analysis_time': 0.0,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.524161277592694,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.524161277592694,
'retro_star_rxn_cost': 4.524161277592694}

Reaction

{'analysis_time': 0.165979,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 4.524161277592694,
'retro_star_reaction_number': 4.524161277592694,
'retro_star_rxn_cost': 0.03971888849948353,
'retro_star_value': 4.524161277592694}



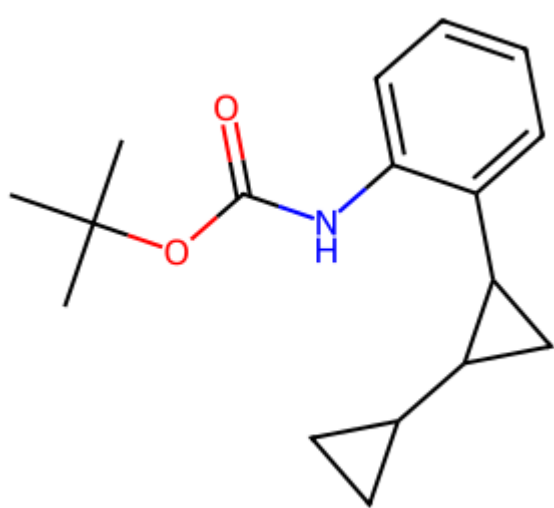
{'analysis_time': 0.165979,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.484442389093211,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.484442389093211,
'retro_star_value': 4.524161277592694}



{'analysis_time': 0.165979,
'first_solution_time': 0.165979,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 4.524161277592694}

Reaction

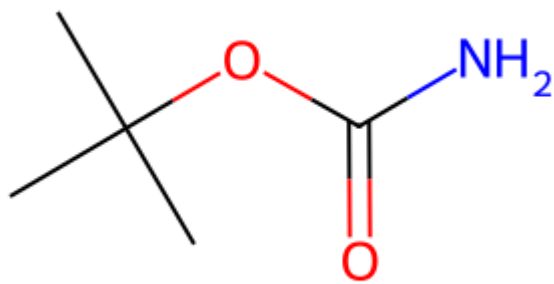
{'analysis_time': 0.26065,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 2,
'num_calls_value_function': 64,
'retro_star_min_cost': 4.484442389093211,
'retro_star_reaction_number': 4.484442389093211,
'retro_star_rxn_cost': 0.15411857871950294,
'retro_star_value': 4.524161277592694}



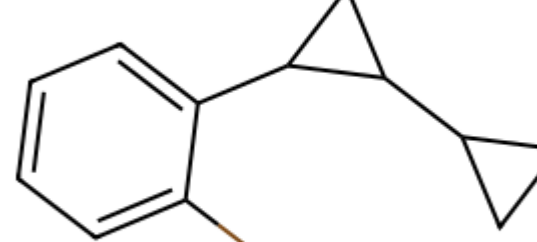
{'analysis_time': 0.26065,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 2,
'num_calls_value_function': 64,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.330323810373708,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.330323810373708,
'retro_star_value': 4.524161277592694}

Reaction

{'analysis_time': 0.375322,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 3,
'num_calls_value_function': 107,
'retro_star_min_cost': 4.330323810373708,
'retro_star_reaction_number': 4.330323810373708,
'retro_star_rxn_cost': 1.2690639113893676,
'retro_star_value': 4.524161277592694}



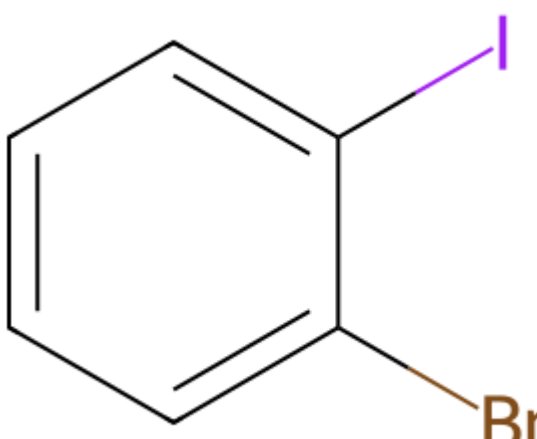
{'analysis_time': 0.375322,
'first_solution_time': 0.375322,
'num_calls_rxn_model': 3,
'num_calls_value_function': 107,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 4.524161277592694}



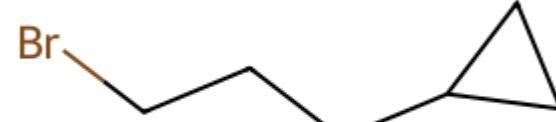
{'analysis_time': 0.375322,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 3,
'num_calls_value_function': 107,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.0612598989843405,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.0612598989843405,
'retro_star_value': 4.524161277592694}

Reaction

{'analysis_time': 0.485153,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 4,
'num_calls_value_function': 170,
'retro_star_min_cost': 5.534371141368224,
'retro_star_reaction_number': 5.534371141368224,
'retro_star_rxn_cost': 5.534371141368224,
'retro_star_value': 6.997272519976578}



{'analysis_time': 0.485153,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 4,
'num_calls_value_function': 170,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 6.997272519976578}



{'analysis_time': 0.485153,
'first_solution_time': 0.485153,
'num_calls_rxn_model': 4,
'num_calls_value_function': 170,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 6.997272519976578}